GAN-based Radar Micro-Doppler Augmentation for High Accuracy Fall Detection System

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***Abstract*—Convolution Neural Network (CNN) is one of the powerful deep learning tools used in many computer vision tasks; however, it is still in a premature state while dealing with sensor data due to the unavailability of a large data set. Here in this paper, we present a CNN-based fall detection system. Deep convolution generative adversarial network (DCGAN) is used to synthesize more fall data before analysis. Sensor data along with synthesized data is prepossessed, and time-varying spectrograms are derived. We apply CNN on raw spectrogram images of different activities to categorize a fall from other activities. As a result, we successfully attain a classification accuracy of 97.2%, surpassing the conventional machine learning methods and previous works on the same data by a large margin.**

***Index Terms*—CNN, FMCW Radar Sensor, GAN, Human Fall Detection.**

I. INTRODUCTION

According to the World Health Organization (WHO), falls are the second foremost reason for unintended or accidental death worldwide. Each year around 6,46,000 people die from injuries due to falls, and 80% of those deaths are from low and middle-income countries. Although older adults are more prone to fall, young adults and children are also not barred from it. The Centers for Disease Control and Prevention (CDC) reported that children aged 19 and under account for 8000 hospital emergency room visits every day for fall-related injuries. Fall-associated injuries may include ache, dysfunctionality, permanent damage to body parts, and sometimes even premature death [1]. Quick detection of the fall followed by a proper response by the concerned authority or caregiver decide the person's well-being.

Various fall detection systems have been proposed starting from wearable systems like accelerometers, push-button types to non-wearable systems like video cameras, radar, etc. [2]. Wearable systems need user interventions, or it has to be carried on the body parts all the time, which is not feasible. Camera-based systems [3] suffer from privacy issues as they will intervene in a person's private life if installed in a bedroom or bathroom-like areas. Furthermore, its performance highly deteriorates in occlusion and low lighting conditions. The radar-based system has emerged as one of the researcher's prominent choices for automatic fall detection [4]. The ability to operate in an adverse environment such as darkness, rain, fog, dirt, heat, etc., helps to build a robust fall detection system. Its small size, strong resistance to radio frequency (RF) interference, and long detection range make it a more suitable choice for fall detection [5]. However, one disadvantage of the radar sensor is that it reacts to any movement. Therefore, we can not easily get the desired activity information if there are other movements in the radar working area.

The occurrence of a fall is a very infrequent and diverse phenomenon, leading to insufficient training data for the classifiers. Varieties of falls can occur, and their unexpectedness makes it even more challenging to design an accurate system [6]. Natural fall data can be collected by installing the devices for an extended period in fall-prone areas like the bathroom and old homage [7]. However, due to difficulties in natural fall data collection, maximum research depends on the artificial fall data collected by performing false fall-like activities in a laboratory setup with the help of young volunteers. Radar systems play a crucial role while collecting data in laboratory setup due to their user-friendly characteristics.

The primary problem researchers face while dealing with sensor data for fall detection is feature extraction [8]. It requires specific domain knowledge to identify the efficient properties of the signal. Moreover, the success of any machine learning algorithms like k- Nearest Neighbor (k-NN), Support Vector Machine (SVM), Sparse Bayesian Classifier, etc., depends upon how efficient our features are [9]. Automatic feature extraction with the help of deep learning makes the task easier [10], [11]. However, it requires a large dataset to increase the layers of the neural networks. Obtaining a sizeable human activity dataset is challenging due to various reasons like human resource limitations. In this case, GAN-based approaches can be beneficial, augmenting synthetic data from actual data.

The main contributions of this study are the following:

- We propose Deep Convolution neural network-based Generative Adversarial Networks (DCGAN) to synthesize more fall-like data that will help us to achieve higher classification accuracy. It can solve one of the major limitations, lack of data, faced by fall detection-based research.
- We also propose a CNN model that takes time-varying raw spectrograms as input for classifying a fall from other non-fall activities. Direct use of spectrogram images enables us to perform classification without getting involved

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in manual feature extraction where domain knowledge is necessary.

II. LITERATURE SURVEY ON SIMILAR WORK

Fall detection methodologies has several aspects that need special attention. One such aspect is to decide the position of the sensor [12]. Deep learning based models require a large amount of data for higher accuracy [13]. In human motion analysis, the large amount of data collection involves a higher cost which is an inherent problem. As a solution, Alnujaim et al. proposed Generative Adversarial Networks (GAN) to augment human motion micro-doppler data. Total seven human activities were collected using a micro-Doppler sensor. Due to the use of GAN, the accuracy of human motion recognition increased from 90% to 94% for the same CNN model that was used before using GAN [14]. Alharbi et al. proposed Wasserstein GAN (WGAN) to synthesize balanced and realistic sensor data automatically. Synthetic data were assessed using CNN and the Long Short-Term Memory (LSTM) network. Accuracy was improved for incorporating WGAN synthesized data with the real data [15]. Kim et al. also proposed a CNN model to identify humans from other objects like dogs, horses, and cars. Seven different human activities were also performed for data collection. The proposed method achieved an accuracy of 97.6% for human detection, and 90.9% accuracy for human activity classification [16]. Li et al. suggested a Bidirectional LSTM (bi-LSTM) network that implemented the fusion of features of both wearable and radar sensors that attained an accuracy of 96 % [17]. In our work, we use DCGAN along with the CNN model to classify fall and non-fall activities. Accuracy has been increased from 96% to 97.2% for the same dataset [18]. Summary of previous and our proposed work on radar based fall detection system has been presented in table I.

III. PROPOSED WORKFLOW

The proposed workflow is presented with a block diagram as shown in Fig1. In this work, we are using open-source data that were collected at the University of Glasgow using a Frequency Modulated Continuous Wave radar sensor (FMCW) operating at 5.8 GHz center frequency and an instantaneous bandwidth of 400 MHz [18]. For convenience now on, we refer to open data as Glasgow data. All the Radar specifications are tabulated in the Table II. A total of six types of human activities were recorded by sixty volunteers aged between 21 to 98, such as walking, sitting down, stand up, pick up an object, drink water, and fall. In this work, we focus on fall recognition, where our goal is to discriminate fall from all the other five activities. At first, we generated synthetic fall data using GAN to increase our data. Then, time-varying micro-doppler signatures were extracted from the different activity data to use as a input in the classification model. 80% of the total data was used to train the model, and 20% were used for testing purposes.

A. Data collection

The used FMCW radar sensor architecture is shown in Fig. 2. At first, a synthesizer generates an electromagnetic wave with the varying frequency with time, also called a chirp. Then the generated electromagnetic(EM) wave is transmitted through the transmitter antenna, and after hitting the target, a few portions are reflected and received at the receiver antenna. The received signal passes through the mixer and results in IF(Intermediate Frequency) signal, which gets amplified before passing through a low pass filter (LPF) and an analog to digital converter (ADC) [34]. After converting into the digital domain, further data processing takes place. Fioranelli et al. collected most parts of the data at the University of Glasgow. A few portions were collected at North Glasgow Housing Association facilities and Age UK West Cumbria Centre [18]. Volunteers were encouraged to do few repetitions of each of the six activities, which includes walking back and forth from the sensor, sitting down on a chair, standing up from the chair, bending forward to pick up an object, drinking anything from the cup or glass and a frontal fall. Fall data were collected in a laboratory-controlled environment with only physically fit volunteers. Walking data have a duration of 10s, and all other activities have a span of 5s.

B. Generative Adversarial Networks(GAN)

GAN is an unsupervised machine learning technique that discovers the patterns and regularities in the original data so that it can produce a similar kind of synthesized data. It has a wide range of applications due to its unique capabilities, ranging from image de-noising to image-to-image translation [35]. A GAN is based on the game theory concept of minimax game, and it is made up of two sub-models, such as a generator model (G) and a discriminator model(D). These two sub-models compete with each other at the time of training to optimize the outcome of the system. The generator model generates new plausible data, and the discriminator model decides whether the data is generated or taken from the actual sample. The discriminator uses a binary classification approach using a sigmoid function resulting in the output value in the range of 0 to 1. The generative model analyzes



Fig. 1: Proposed Workflow

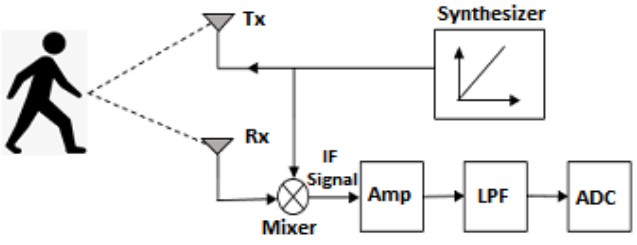


Fig. 2: FMCW Radar Sensor Architecture

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TABLE I. **Previous Work on Radar-based Fall Detection System**

Reference	Radar Type (Operating Frequency)	Dataset	Feature Extraction Method	Classification Method (Accuracy)
[4]	CW (24GHz)	Fall-121, Non-Fall-117	MPA using Sliding root mean square method	SVM(89%), NB(81%), 5-NN(79%), LDA(88%) and DT (77%)
[19]	IR-UWB(5.9-10.3GHz)	Fall-187, Non-Fall-149	Time frequency plot	CNN(89.12%)
[20]	IR-UWB(5.9-10.3GHz)	Fall-121, Non-fall-85	Time series plot	DCNN(95.83%)
[21]	FMCW(76-81GHz)	Fall + Non-Fall data = 231	STFT Spectrogram	Line Kernal CNN(95.24%)
[17]	FMCW(5.8GHz)+ 3 wearable sensors	Fall- 48, Non-Fall-96	Statistical and time-frequency features	Bi-LSTM(95.9%)
[22]	IR-UWB(5.9-10.3GHz)	Fall-1870, Non-Fall-1490	Time-Frequency Plot	Capsul Network(95.43%)
[23]	CW (10.525GHz)	Fall-500, Non-Fall-250	STFT Spectrogram	CNN(96%)
[24]	CW(8GHz)	Fall-32, Non-Fall-32	Time Frequency Representations	SVM (NA)
[25]	SFCW(5.8GHz)	Fall-60, Non-Fall-40	STFT Spectrogram	DTW(94.3%)
[26]	CW(25GHz)	Fall-120, Non-Fall-1236	STFT Spectrogram	ACGAN-DCNN(82.56%)
[27]	Kinect(4GHz)	Fall-120, Non-Fall-1236	STFT Spectrogram	DivNet(95.7%)
[28]	RCR	Fall-105, Non-Fall-704	Wavelet Transform	1-NN(83%)
[29]	CW(6GHz)	Fall-15, Non-Fall-45	STFT Spectrogram	DNN(87.0%)
[30]	IR-UWB(24GHz)	Fall-80, Non-fall-240	DWT	SVM(79.6%)
[31]	FMCW(2.425GHz)	NA	STFT Spectrogram	CNN(95%)
[32]	CW(8GHz)	Fall-20, Non-Fall-10	Wavelet Transform	Mahalanobis Distance Metric(NA)
[33]	IR-UWB(4GHz)	Fall-102, Non-Fall-408	Wavelet Transform	CNN(92.6%)
Proposed	FMCW(5.8GHz)	Fall-198, Non-Fall-350	STFT Spectrogram	DCGAN+CNN(97.2%)

IR-UWB=Impulse-radio Ultrawide Band, CW = Continuous Wave, FMCW = Frequency Modulated Continuous Wave, SFCW = Step Frequency Continuous Wave, RCR = Range-Controlled Radar, MPA = Major Physical Activities, STFT = Short-time Fourier transform, DWT = Discrete Wavelet transform, SVM = Support Vector Machine, NB = Naive Bayes, k-NN = k-Nearest Neighbour, LDA = Linear Discriminant Analysis, DT = Decision Tree, CNN = Convolution Neural Network, DCNN = Deep Convolution Neural Network, Bi-LSTM = Bidirectional Long Short Term Memory, DTW = Dynamic Time Wrapping, ACGAN = Auxiliary Classifier Generative Adversarial Network, DivNet = Diversity Networks

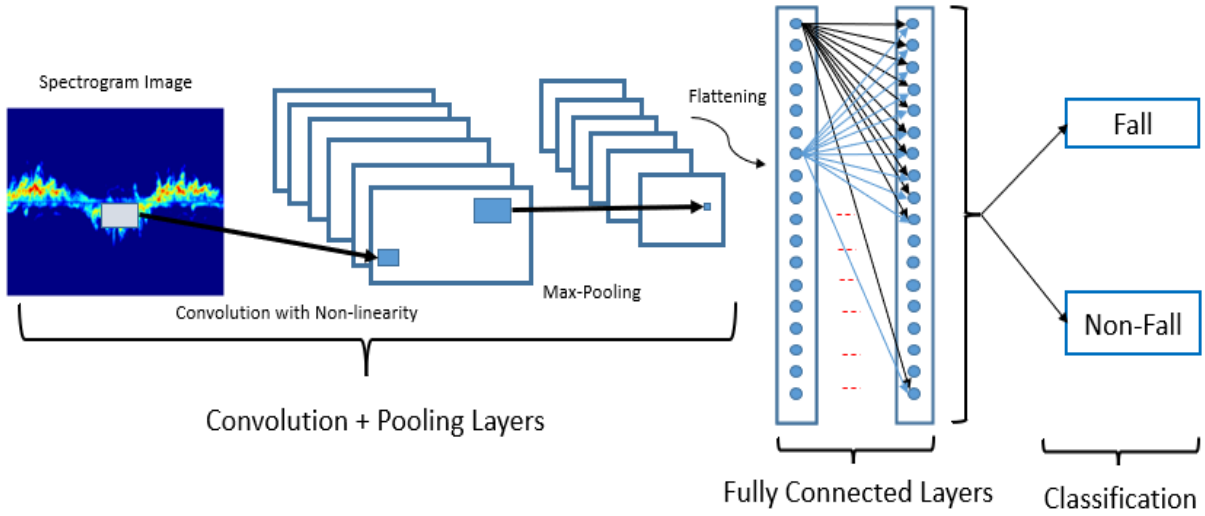


Fig. 3: Proposed CNN Architecture

TABLE II. Specification of used system

Parameter	Value
Transmitted Power	+19 dBm
Starting Frequency	5.8 GHz
Stop Frequency	6.2 GHz
Bandwidth	400 MHz
Frequency Slope	0.4 MHz/μs
Chirp Duration	1ms
ADC sampling rate	128 KHz

the distribution of the data in such a way that, after the training phase, the probability of the discriminator making a mistake maximizes [36]. Algorithm 1 shows the overall structure of our GAN-based model.

In our case, we have 198 fall data and 350 non-fall data consisting of different activities. Each fall data is five seconds span data, sampled at the rate of 128 kHz, which gives us 6,40,000 samples per data. Direct application of this data into any GAN network would require more powerful processors. In order to successfully apply GAN to these data, we divided it into 100 parts of 6,400 samples each. We train all the 100 samples independently; then, we combine all the generated 100 samples to get one complete five-second fall data. We

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Algorithm 1 Generalised Adversarial Network (GAN)

1) **Define the Problem**
   At first, the goal is set, and data is collected.
2) **Set Architecture of the GAN**
   Based on the specific problem, we set the architecture of the GAN. In our case, the generator model consists of four convolutional layers followed by ReLU activation. Furthermore, the final convolutional layer has activation of Tanh. In contrast, the discriminator model consists of five convolutional layers followed by Leaky ReLU activation. Furthermore, The final convolutional layer has a sigmoid activation. Batch Normalization is used to avoid overfitting. The batch size is kept to 16, and the learning rate is 0.002. Adam optimizer is used with a beta value of 0.5. The number of input channels to the training set data is 2, and the size of feature maps is 6,400.
3) **Train the Discriminator with real data**
   Train the discriminator with the real data to predict them as a real value.
4) **Generate Synthetic Data from Generator**
   In this step, we generate fake or synthetic samples out of the generator.
5) **Train the Discriminator with Synthetic Data**
   Again train the discriminator with the fake data generated by the generator. This way discriminator knows what kind of data is accurate and which is fake.
6) **Train Generator with the help of Discriminator Output**
   Based on the predictions, we train the generator to fool the discriminator. The generator's (G) objective is to generate data indistinguishable from the actual data by discriminator (D).

consider the deep convolutional neural network-based GAN (DCGAN). DCGAN is nothing but uses convolutional layers in the generator and discriminator models. We successfully increased our fall data from 198 to 250 with the help of the GAN.

C. Preprocessing

Fig. 3 shows a blueprint of our proposed CNN model. Before feeding the model, data must go through the preprocessing part to remove unwanted noise and components often created from the small vibrations of the device. The necessary preprocessing steps are shown in Fig 4. At first, the data was rearranged in a matrix form containing 128 rows and M columns, where M represents a total number of chirps in the activity duration and 128 is the number of time samples within one chirp. Then, we apply fast Fourier transform (FFT) to each column of the data matrix and find spectrum centered at the beat frequency, which is proportional to the target's distance from the sensor. Thus after obtaining the range profile of the target, we apply a filter to identify only moving objects. A 9th order infinite impulse response (IIR) high-pass Butterworth



Fig. 4: Preprocessing Steps

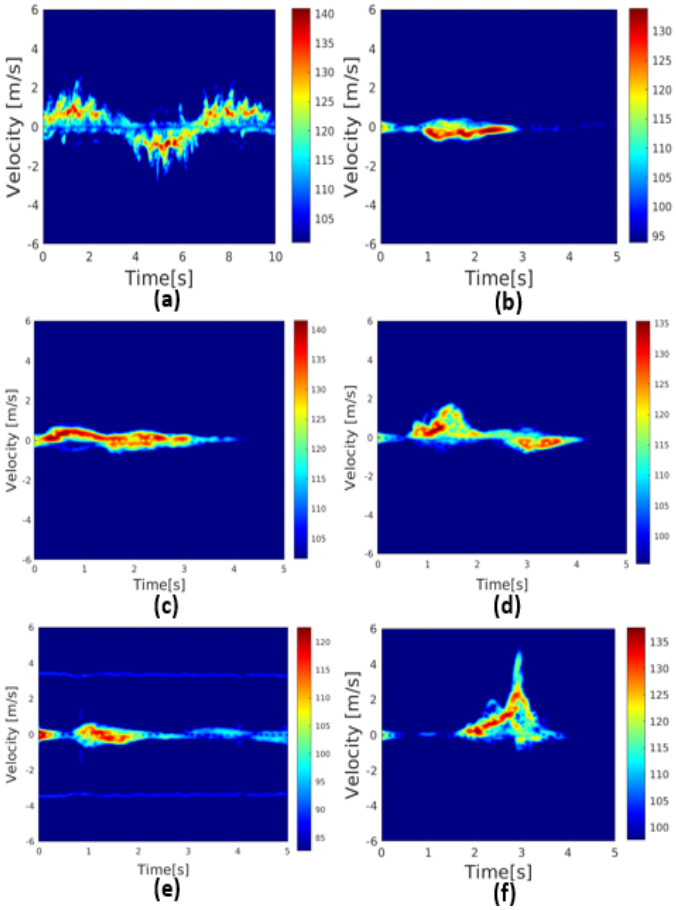


Fig. 5: Time-varying spectrograms of six actions: (a) Walking; (b) Sitting Down; (c) Stand up; (d) Pick up an object; (e) Drink Water; (f) Fall

notch filter having a cut-off frequency 0.0075Hz was applied. It helps us remove the range components near the 0Hz frequencies created by stationary objects surrounded by the target. Then Short-time Fourier transform (STFT) was implemented, having a 0.2s hamming window and 95% overlapping factor on the obtained range-time data matrix to extract the time-varying micro-doppler spectrogram [37]. Fig. 5 shows the spectrogram images of different activities.

D. Feature Extraction

Human movement generates two kinds of signatures, i.e., Doppler and micro-Doppler. Doppler signature is based on the movement of the overall body, where micro-Doppler signatures are generated due to the movement of the different limbs of the body, like swinging of head, torso, and limbs of the person. Micro Doppler frequencies represent the sideband of the primary Doppler frequency [38]–[40] and play a prominent role in distinguishing different human activities

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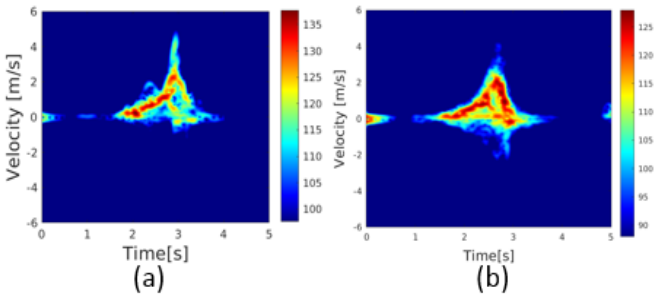


Fig. 6: Time-varying spectrogram of fall (a) without GAN-generated data, (b) with GAN-generated data

with higher accuracy. To incorporate micro-Doppler signatures in our experiment, we use time-varying spectrogram images directly as input to the CNN model. Images were reshaped to a dimension of 54×68 before feeding to the input layer and extracting complex features.

Our proposed CNN model consists of two convolution layers followed by two pooling layers and four fully connected layers. The first convolution layer contains six, and the second convolutions layer consists of sixteen convolution filters of size 5×5 each. We used a maxpool of size 2×2 for pooling layers, and the usual ReLU (Rectified Linear Unit) served as the activation unit. The primary purpose of ReLU is to remove all the negative values from the feature matrix while keeping the positive values intact. Pooling is applied after passing through the activation layer, and it is responsible for shrinking the image stack into a smaller size.

At last, we took our filtered and shrunk images and stack them together in one column, often called flattening of data, to feed into the fully connected layer. In fully connected layers, every neuron of the first layer is connected to every neuron in the second layer. To optimize the weights of the convolution filters backpropagation algorithm used cross-entropy loss and an Adagrad optimizer with a learning rate of 0.0001. We run the optimization loop for 100 epochs, and as the data size was not that big, it took around five minutes to train the model.

E. Classification

Softmax function, also known as a normalized exponential function, was used as the activation function in the last layer of the model to normalize the output to a probability distribution over predicted output classes.

IV. EXPERIMENTAL RESULTS

We generated a total of 52 fall data using GAN. Fig 6 shows a time-varying spectrogram of augmented fall data along with a spectrogram of original fall data . We used five-fold cross-validation to find the effectiveness of our model. At first, we divided our whole dataset into five equal parts. Then we train the model with four folds and test the model using the remaining one fold. This same process is repeated until each fold serves as a test set once.

A. Results of Our Approach

We note down the accuracy while each fold acts as a test set and show in Table III. We can see from the table, we have achieved the highest accuracy of 99% using fold-1 as the test set and the lowest accuracy of 95% using fold-4 as the test set. Overall, an average of 97.2% accuracy is attained by our proposed model, which was improved by 1.1% for using the GAN generated fall data along with the real fall data as shown in Table IV. We also measured the standard deviation of our results to 1.48.

B. Comparison with Previous Approaches

We compared our work with previous works [41], [42], [43] and [17] on the same Glasgow dataset [18]. The compared works deal with multi-class classification having fall one of the class. We show the accuracy of only falls vs. other activities from the confusion matrix. The accuracy of all the mentioned work is shown in Table V. It is clearly seen that, our proposed method achieved a higher accuracy as compared to others. Our proposed DCGAN structure generates high-quality synthetic data, which eventually increases the count of our training data, leading to a higher accuracy.

TABLE III. Five-fold cross validation

Fold	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Accuracy	99%	97%	98%	95%	97%

TABLE IV. Accuracy With and Without synthetic data

Without synthetic data	96.1%
With synthetic data	**97.2%**

TABLE V. Comparative Analysis on Glasgow data [18]

Ref.	Method	Accuracy
[41]	Sensor Fusion	91.3%
[42]	Spectrogram features with different classifiers	94%
[43]	SVM+Artificial Neural Network (ANN)	96%
[17]	Bi-LSTM Network	95.9%
Proposed	**DCGAN+CNN**	**97.2%**

V. CONCLUSION AND FUTURE WORK

CNN and other deep learning techniques have received close attention in various research areas. However, collecting a large dataset for training the neural network is still a big challenge. This paper presented Generative Adversarial Network to augment Doppler radar data of human falls measured by the FMCW radar setup. We used CNN for the classification of a human fall from other activities. As micro-Doppler characteristic provides minute signatures of particular human activity, we prefer raw time-varying spectrogram as the input of the CNN model. One point to be noted here is that in this work, we have not used much domain knowledge to extract the complex features out of the data; instead, the spectrogram itself served as an input to our CNN model. Due to the augmentation of data, we could construct and train deeper neural networks effectively, and as a result, we achieved a classification accuracy of 97.2%. We want to shift our research work from binary to multi-class classification in the future.

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